

Ridwan Sharif

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SUMMARY

Experienced with distributed systems, databases, operating systems, infrastructure, performance, and backend engineering

EDUCATION

University of Waterloo

Sept '16 – Apr '21

Bachelors of Computer Science, Honours (with Distinction)

WORK EXPERIENCE

Google

Toronto, ON

Senior Software Engineer (Cloud Observability)

June '21 – Present

- Engineering lead of the managed observability teams, enabling over **\$150M+** in ARR
- Enabled metrics collection for Cloud Run, observing **1M+** serverless workloads
- Architected scalable telemetry collection, storage, and routing for GKE, GCE, Cloud Run, etc.
- OSS Core Contributor to OpenTelemetry and Prometheus, multiple time speaker at KubeCon ([1],[2])

Google

Sunnyvale, CA

Backend Engineering Intern (gVisor)

May – Sep '20

- Worked on expanding the gVisor kernel by supporting Linux compatible sandboxed FUSE filesystems
- Built the associated kernel module, device drivers, filesystem syscalls, and protocol implementation in gVisor

Postmates / Uber

San Francisco, CA

Platform Engineering Intern (Delivery as a Service)

Jan – Apr '20

- Built the DaaS invoicing and billing system, reducing unrecoverable debt by **\$50,000+** annually
- Architected the debt collection service by integrating with BigQuery, GCS, Square, Shopify, Stripe

Cockroach Labs

New York, NY

Backend Engineering Intern (SQL Optimizer)

Sep '18 – Aug '19

- Added support for multi-table mutations, ordered aggregations, segmented sort and look-up joins
- Implemented index-skip-scans to improve index scan performance by an order of magnitude

Backend Engineering Intern (Distributed SQL)

- Designed and implemented load based splitting to replicate and distribute ranges of data across nodes
- Improved overall read/write throughput by over **60%** and **halved** end-to-end latency

Shopify

Ottawa, ON

Production Engineering Intern (Infrastructure)

Jan – Apr '18

- Led multiple engineering initiatives outlining measures to improve observability across the platform
- Built incident report and cluster health tooling that was featured on the company engineering blog

RESEARCH

uWaterloo Database Systems Group

2018

Undergraduate Researcher (Prof. Grant Weddell)

Waterloo, ON

- Researched main-memory databases that use dynamic code-generation as part of its execution model

PROJECTS

- **raft** – A performant distributed consensus algorithm; based on Diego Ongaro's thesis paper
- **mqueue** – Constructed an in-memory Kafka-esque message broker in Go over an HTTP API